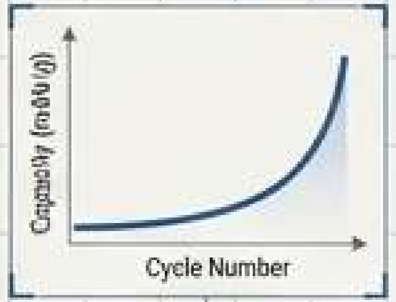
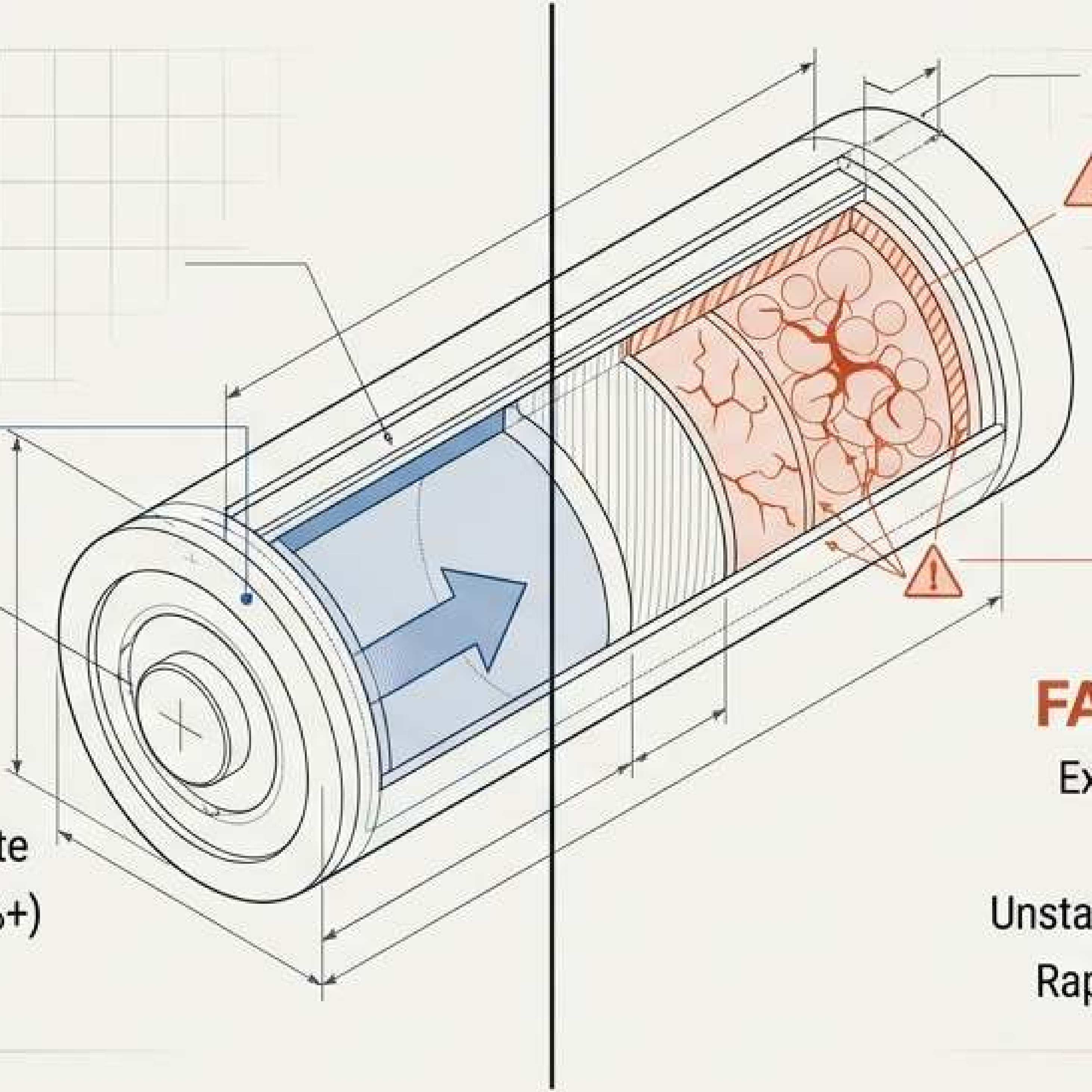
SILICON ANODE BATTERIES

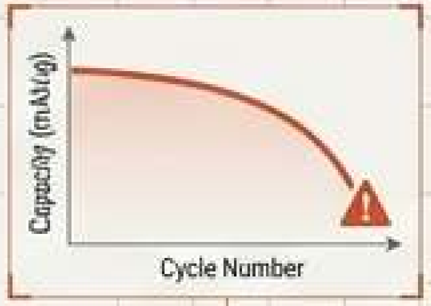
The Holy Grail of Energy Density and the Price of Unlocking It

PERFORMANCE PROMISE





ENGINEERING RISK



ENERGY DENSITY BREAKTHROUGH

- 10x Theoretical Capacity of Graphite
- Significantly Increased Range (50%+)
- Faster Charging Potential

FAILURE MODES & RISKS

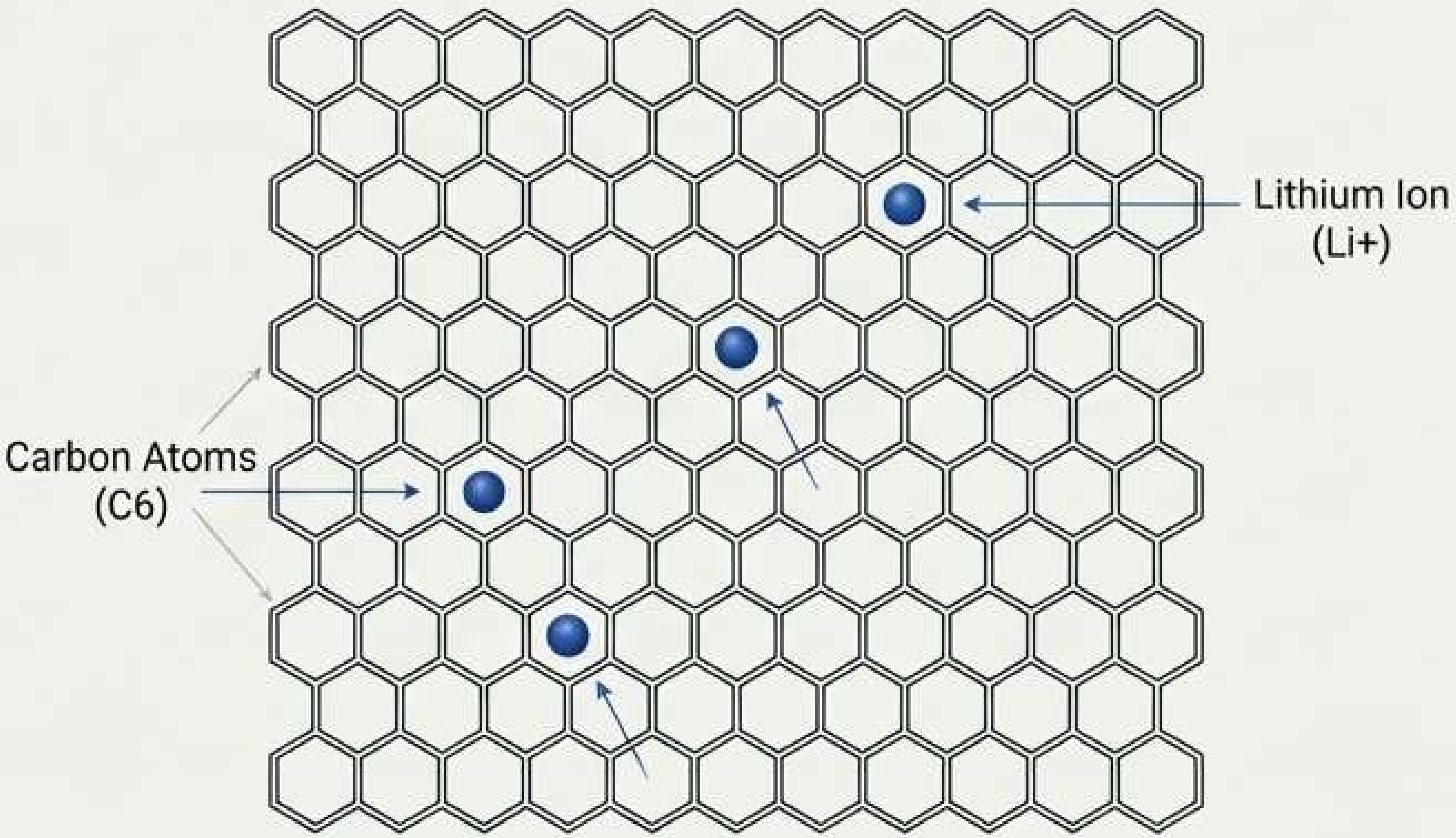
- Extreme Volumetric Expansion (300%)
- Electrode Pulverization & Cracking
- Unstable Solid Electrolyte Interphase (SEI)
- Rapid Capacity Fade & Safety Concerns

THE HARD CEILING OF LEGACY LITHIUM-ION

GRAPHITE LIMITATIONS

- **Graphite** has served as the standard commercial anode since the 1980s due to its **structural stability**.
- The **physical limit**: It requires **six carbon atoms** to hold a **single lithium ion**.
- Theoretical maximum capacity of **372 mAh/g**.

APARTMENT VS. SPONGE (PART 1): THE RIGID GRAPHITE LATTICE



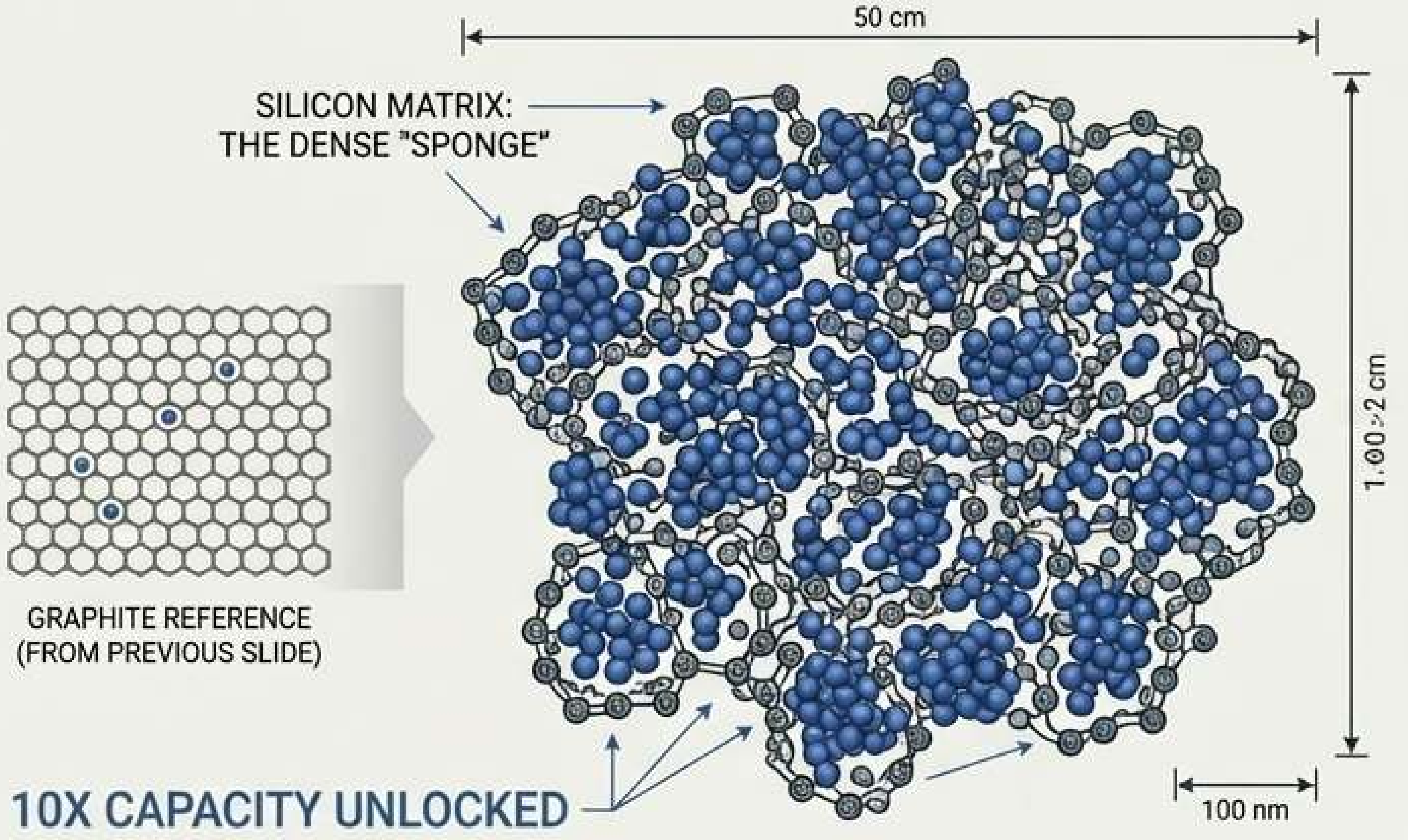
RIGID & LIMITED CAPACITY: Only one Li+ fits per C6 unit.

THE MATHEMATICAL SUPERIORITY OF SILICON

SILICON ADVANTAGES

- Silicon bonds with lithium differently, accommodating **3.75 lithium ions** per silicon atom.
- This chemical reality unlocks **10x** the **gravimetric capacity** and **3x** the **volumetric capacity** of graphite.
- Near-theoretical capacity of **3569 mAh/g**.

APARTMENT VS. SPONGE (PART 2): THE SILICON CAPACITY LEAP



DENSE & SATURATED CAPACITY: Multiple Li+ ions accommodated per Si atom, creating a massive energy reservoir.

0 kmh 300

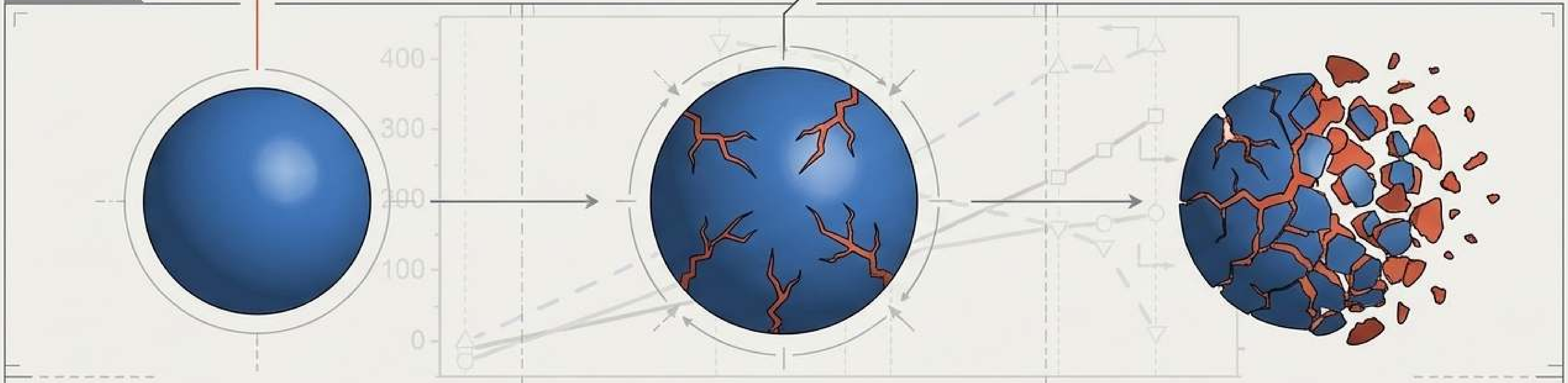
NotebookLM

THE SWELLING PARADOX

Silicon’s ability to absorb massive amounts of lithium causes catastrophic physical expansion.

Volume expansion **>300%** during lithiation.

In conventional designs, this severe swelling pulverizes the anode material and destroys the battery from the inside out.



STAGE 1:
PRISTINE SILICON PARTICLE

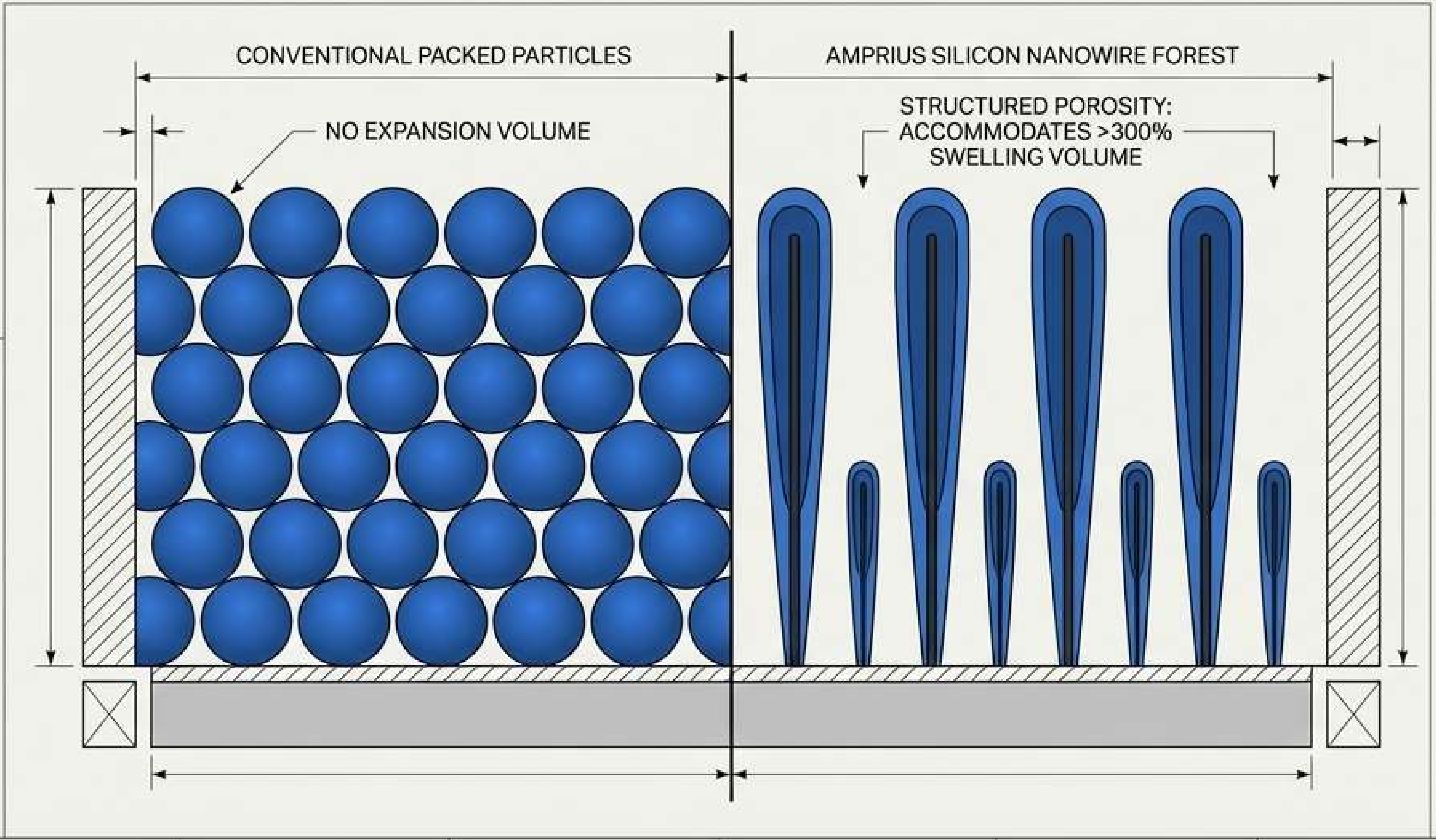
STAGE 2:
LITHIATED & EXPANDED
(STRESS & FRACTURE)

STAGE 3:
PULVERIZATION & FAILURE

THE MICRO-ARCHITECTURE BREAKTHROUGH

Amprius solved the swelling paradox by completely redesigning the anode's geometry, replacing conventional particles with 100% silicon nanowires.

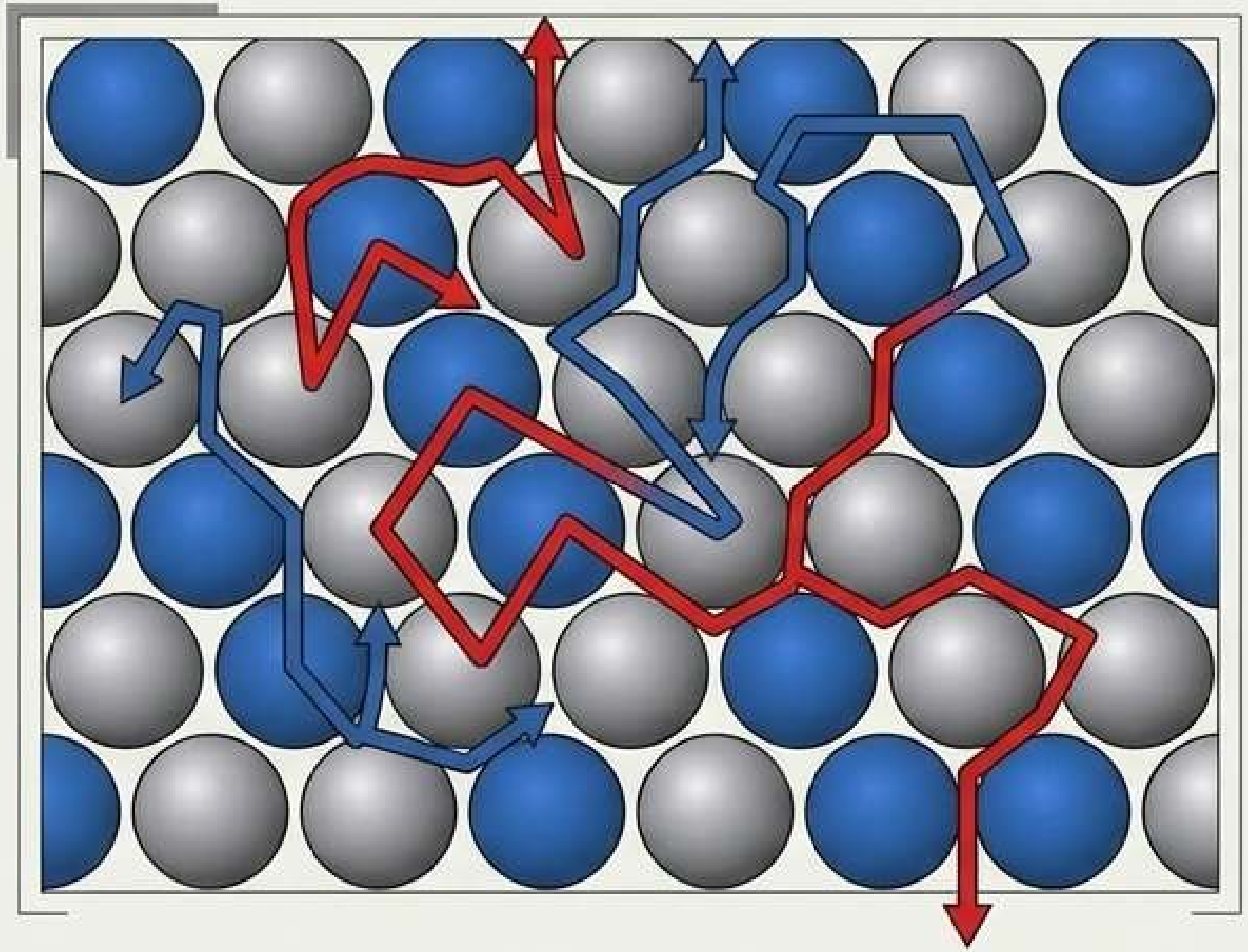
- Structured porosity accommodates >300% expansion.
- Rooted directly to the substrate without binders or inactive materials.



STRAIGHT PATHS FOR EXTREME POWER

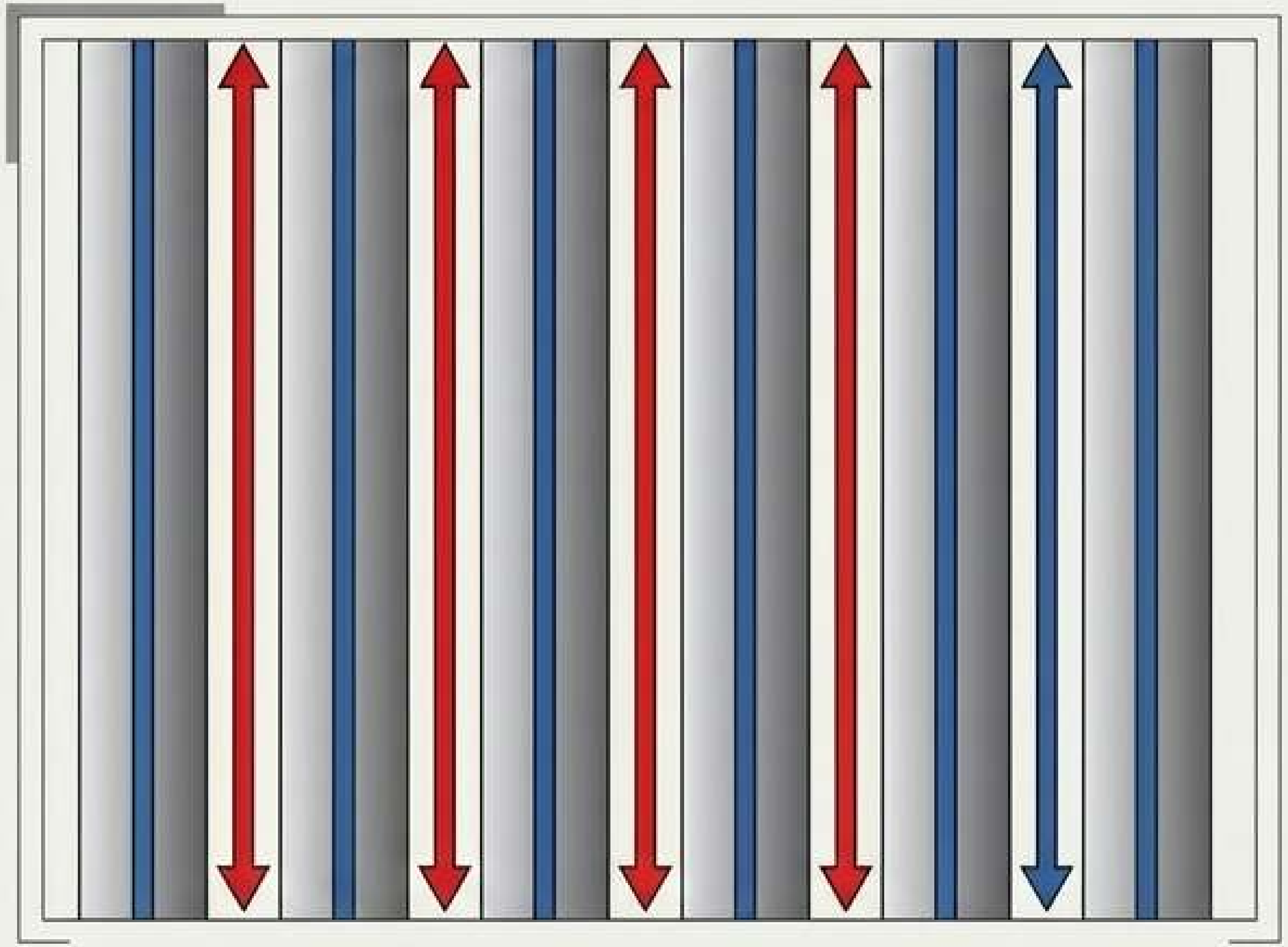
CONVENTIONAL PARTICLE ANODE

Ions and electrons navigate a tortuous, blocked path, slowing charge rates and limiting power.



SILICON NANOWIRE ANODE

Ions and electrons travel straight, direct paths, resulting in high power capability and ultra-fast charging.

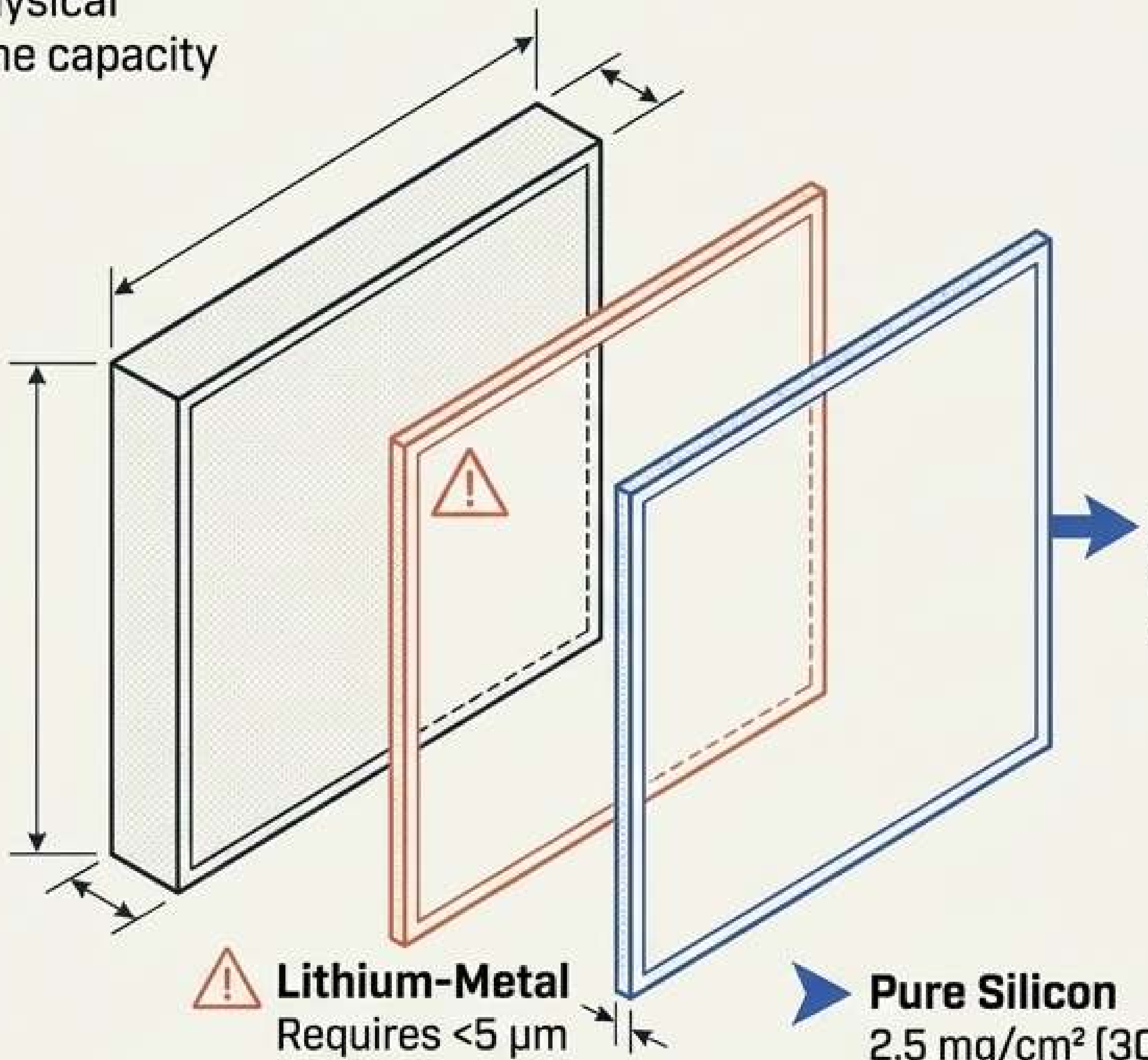


THE SILICON SOLUTION MATRIX

DIMENSION	GRAPHITE (LEGACY)	100% SILICON NANOWIRES (AMPRIUS)
Active Material	Graphite	100% Silicon Nanowires
Volume Expansion Handling	Rigid structure / fails at >10% expansion	Structured porosity spacing absorbs >300% expansion
Ion/Electron Path	Tortuous/Zig-zag	Straight/Direct
Binders & Inactive Materials	Required	Zero required
Theoretical Capacity	372 mAh/g	**3569** mAh/g

THE MASS ADVANTAGE IN CELL DESIGN

Pure silicon radically reduces the physical material required to achieve the same capacity (example based on 5 mAh/cm2).



Conventional Graphite
15 mg/cm2 (96 µm thick)
-> 301 Wh/kg

Pure Silicon
2.5 mg/cm2 (30 µm thick)
-> 407 Wh/kg

Lithium-Metal
Requires <5 µm thickness to compete
-> 344 Wh/kg

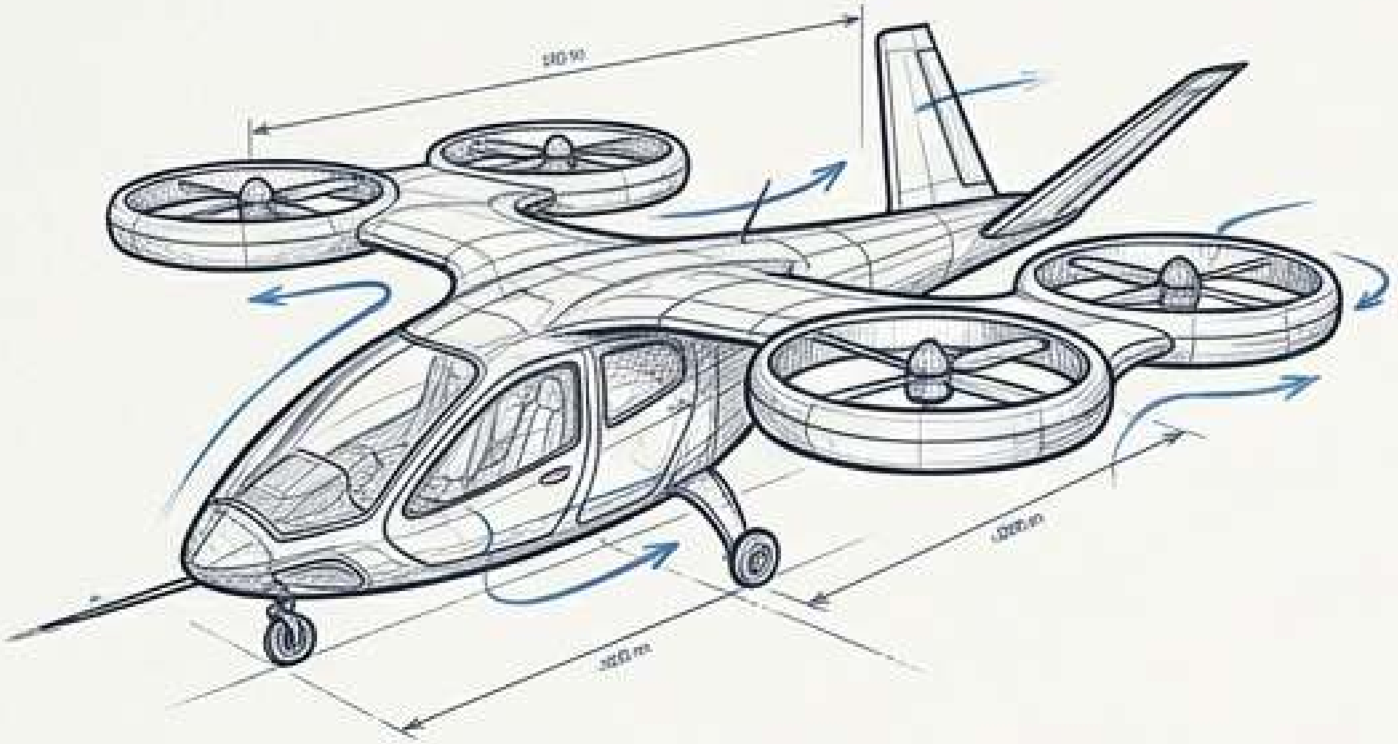
Pure Silicon
2.5 mg/cm² (30 µm thick)
-> 407 Wh/kg

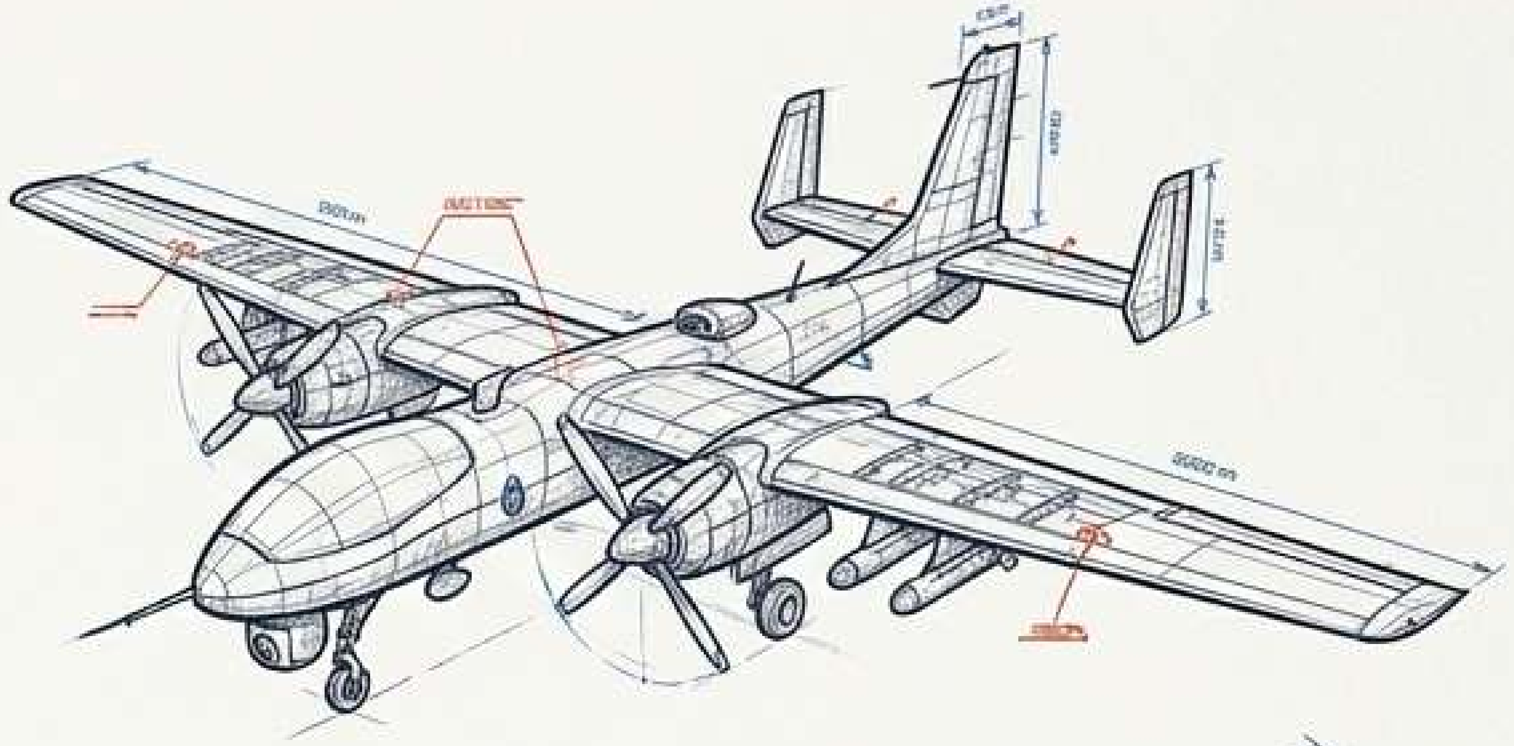
BREAKING THE **ENERGY-POWER TRADE-OFF**

Commercial graphite cells force a compromise between how much energy you carry and how fast you can use it.

Silicon nanowires push the frontier forward, delivering **500 Wh/kg of energy** OR **400 Wh/kg** at massive **4000 W/kg** power density.

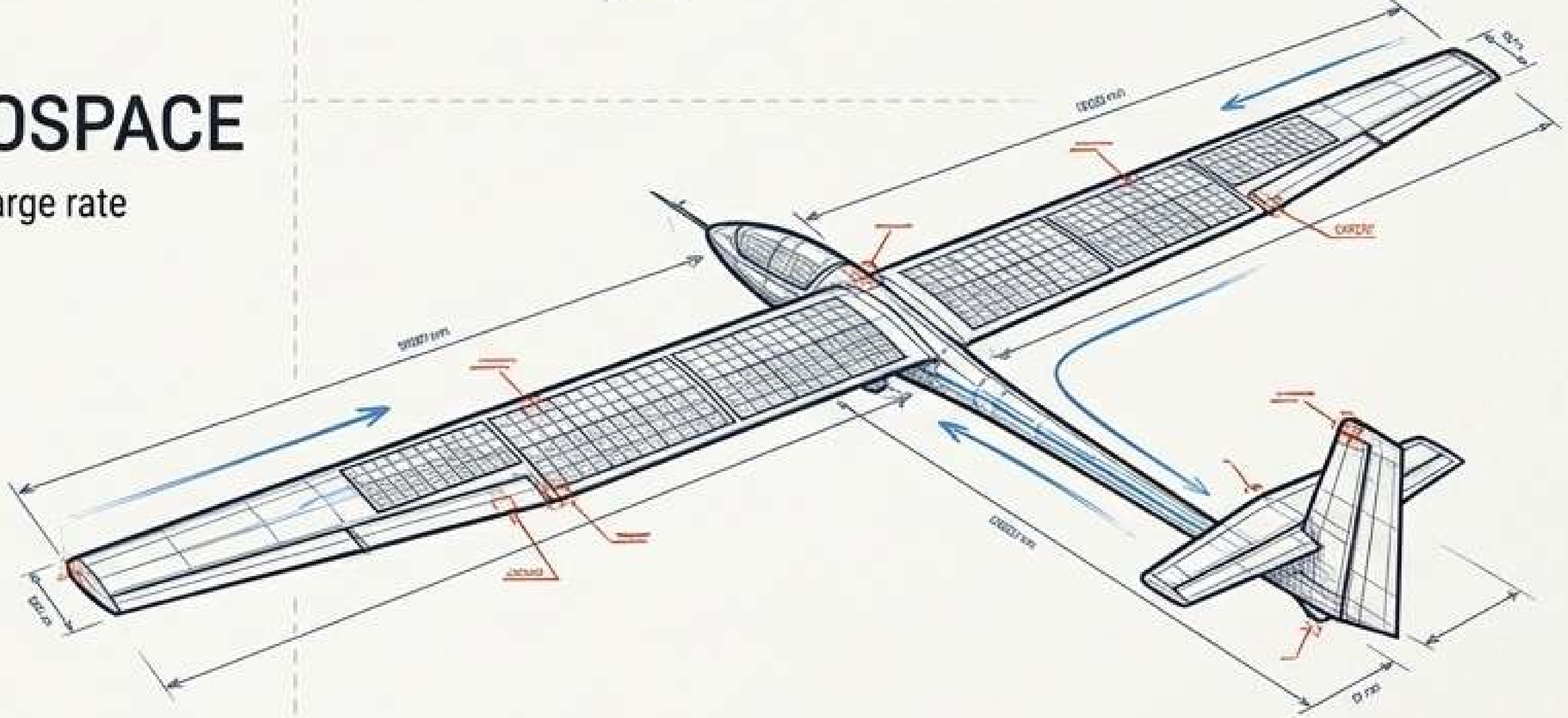
PREMIUM ENGINEERING DOSSIER





UNLOCKING ADVANCED AEROSPACE

This combination of extreme energy density and high discharge rate is the prerequisite for next-generation electric flight.



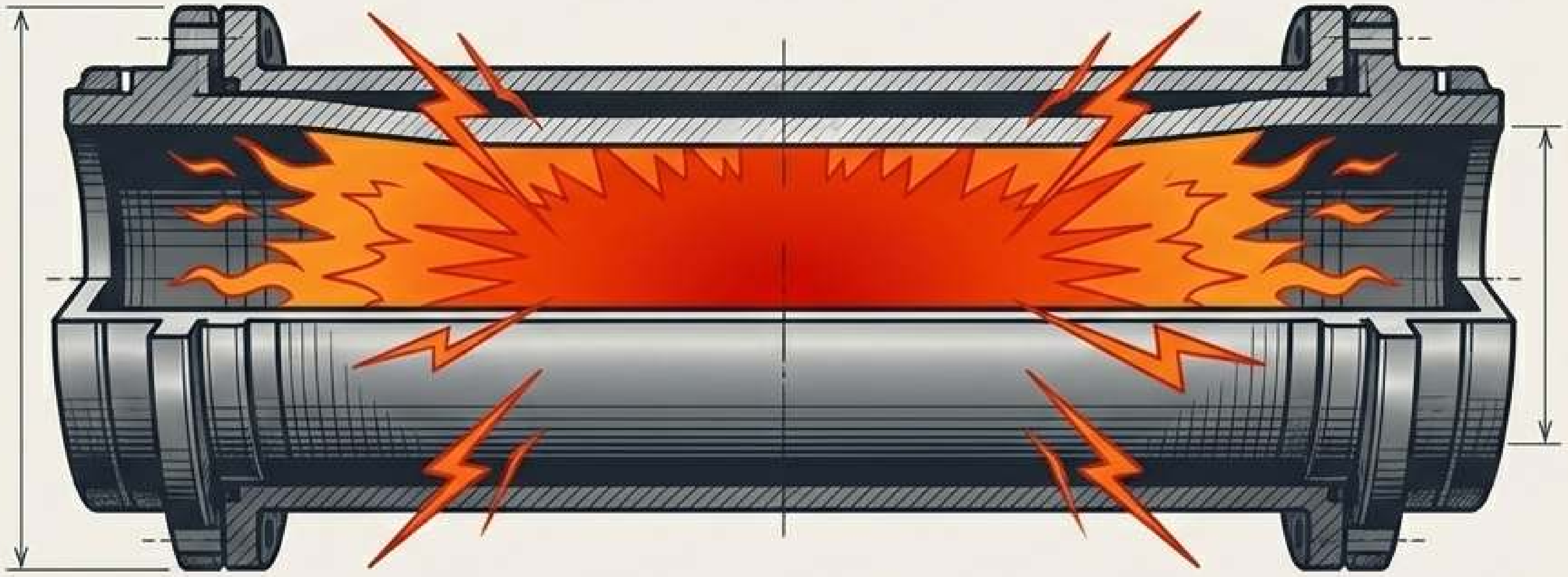
KEY APPLICATIONS:
High-Altitude Pseudo-Satellites (HAPS)
Long-Endurance Drones
eVTOL (Urban Air Mobility)

Operating temp range of -30°C to 55°C allows for atmospheric extremes.

The Reality Check: More Energy, More Risk

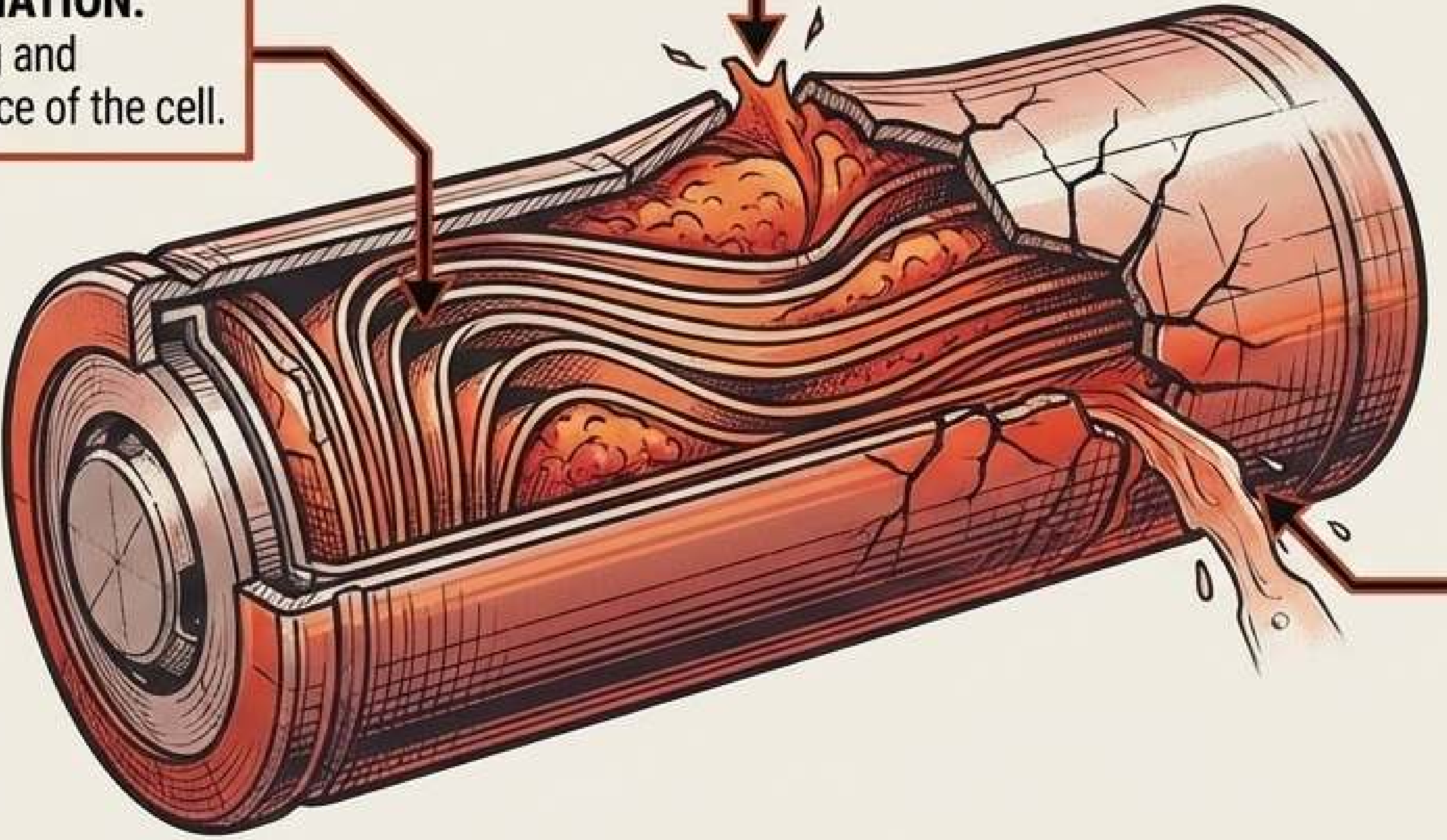
"As the capacity of cells increases, so too does the severity of the thermal runaway event due to the higher energy content." – Exponent

Silicon anodes push the physical limits of energy storage. If a failure occurs, the sudden release of this extreme energy represents a fundamentally **higher danger to devices and consumers than legacy graphite.**



THE ANATOMY OF A SILICON FAILURE

Traditional aging mechanisms can trigger entirely novel failure modes in prototype silicon cells.



The Rush to Commercial Scale

The market for silicon-anode batteries is projected to grow >60% over the next decade. Amprius is scaling aggressively to meet demand.

INTEGRATING WITHOUT DISRUPTION

Amprius utilizes existing commercial manufacturing processes. The only change is the anode manufacturing line. Cathode and assembly lines are completely unchanged.

PROPRIETARY DEPOSITION (100% Silicon Nanowire)

[MIXING] → [COATING] → [CALENDARING] → [SLITTING] → [STACKING] → [FORMATION]

THE NEW PARADIGM FOR SAFETY TESTING

"Slow and steady wins the race." Rushing silicon technology to market risks costly recalls and catastrophic failures.

ABUSE TESTING

Evaluate foreseeable use and extreme misuse scenarios (simulating physical/thermal trauma).

PROPAGATION CONTAINMENT

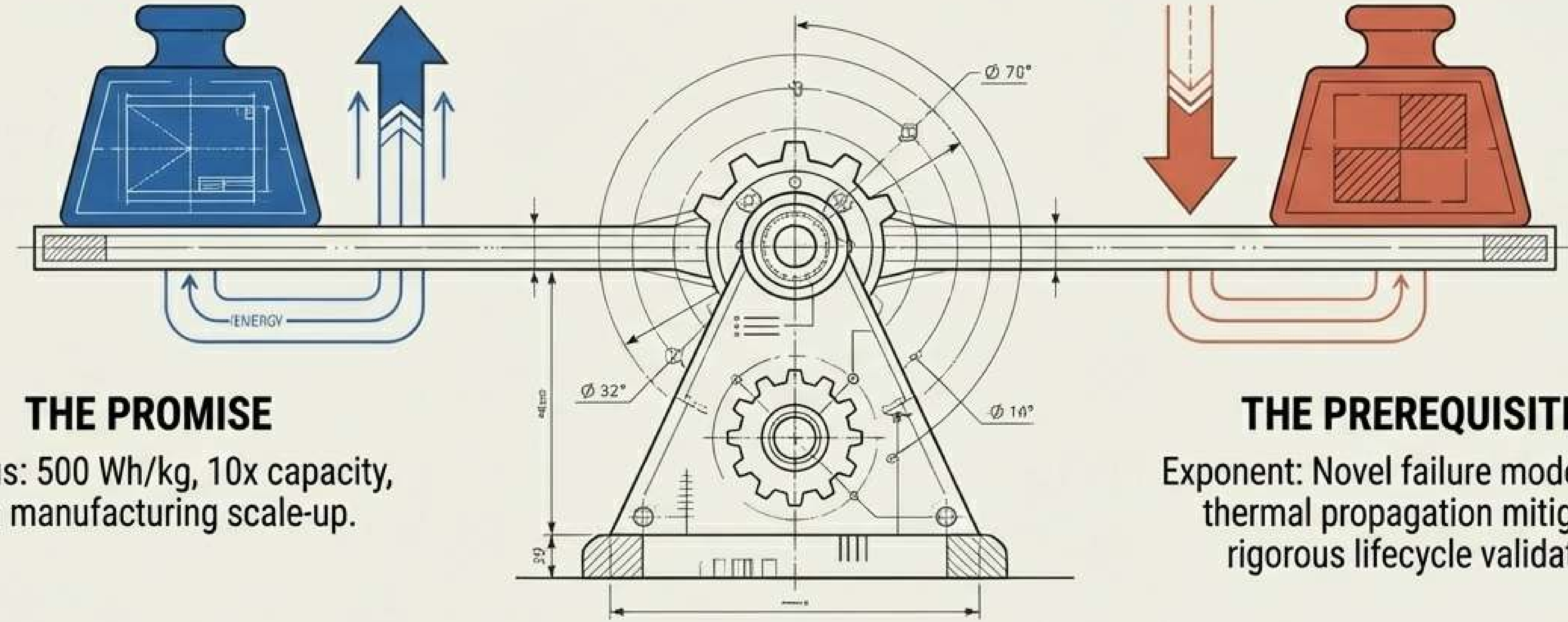
Engineer pack-level mitigation for cell-to-cell thermal propagation (multi-cell pack safety).

LIFECYCLE MONITORING

Develop long-term monitoring for novel aging mechanics (tracking deformation over time).

THE INNOVATION-SAFETY FULCRUM

You cannot scale extreme energy without simultaneously scaling extreme safety validation.
The future of aerospace and mobility depends on balancing both.



THE PROMISE

Amprius: 500 Wh/kg, 10x capacity, GWh manufacturing scale-up.

THE PREREQUISITE

Exponent: Novel failure mode testing, thermal propagation mitigation, rigorous lifecycle validation.